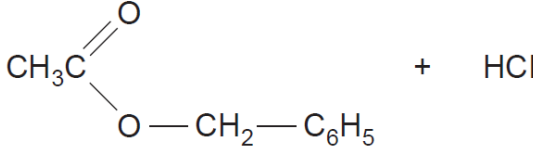


SECTION A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1				 + HCl		1		1		
2	(a)			7.44×10^{14}		1		1	1	
	(b)			it is yellow because the blue end / the rest of the visible spectrum is absorbed	1			1		
3				compound F (1) it contains an O—H bond at $3200\text{--}3500\text{ cm}^{-1}$ and a C=C at 1620 cm^{-1} but no C=O at $1650\text{--}1750\text{ cm}^{-1}$ (1)			2	2		
4				award (1) for any valid formula e.g. $(\text{C}_6\text{H}_5)_3\text{C}\cdot$		1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5				2 signals present (1) as all the <u>C</u> H carbon atoms are equivalent and all the <u>C</u> H ₃ carbon atoms are equivalent (1)			2	2		
6				$\frac{58.0}{(44 + 111)} \times 100 = 37.4$ (2) award (1) if value given to anything other than 3 sig figs		2		2	1	
				Section A total	1	5	4	10	2	0